

POST-FIRE
STRUCTURAL DAMAGE
ASSESSMENT:
2025 LOS ANGELES
PALISADES AND
EATON FIRES

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PROBLEM & MOTIVATION

RAPID SPATIAL PREDICTION OF STRUCTURAL DAMAGE

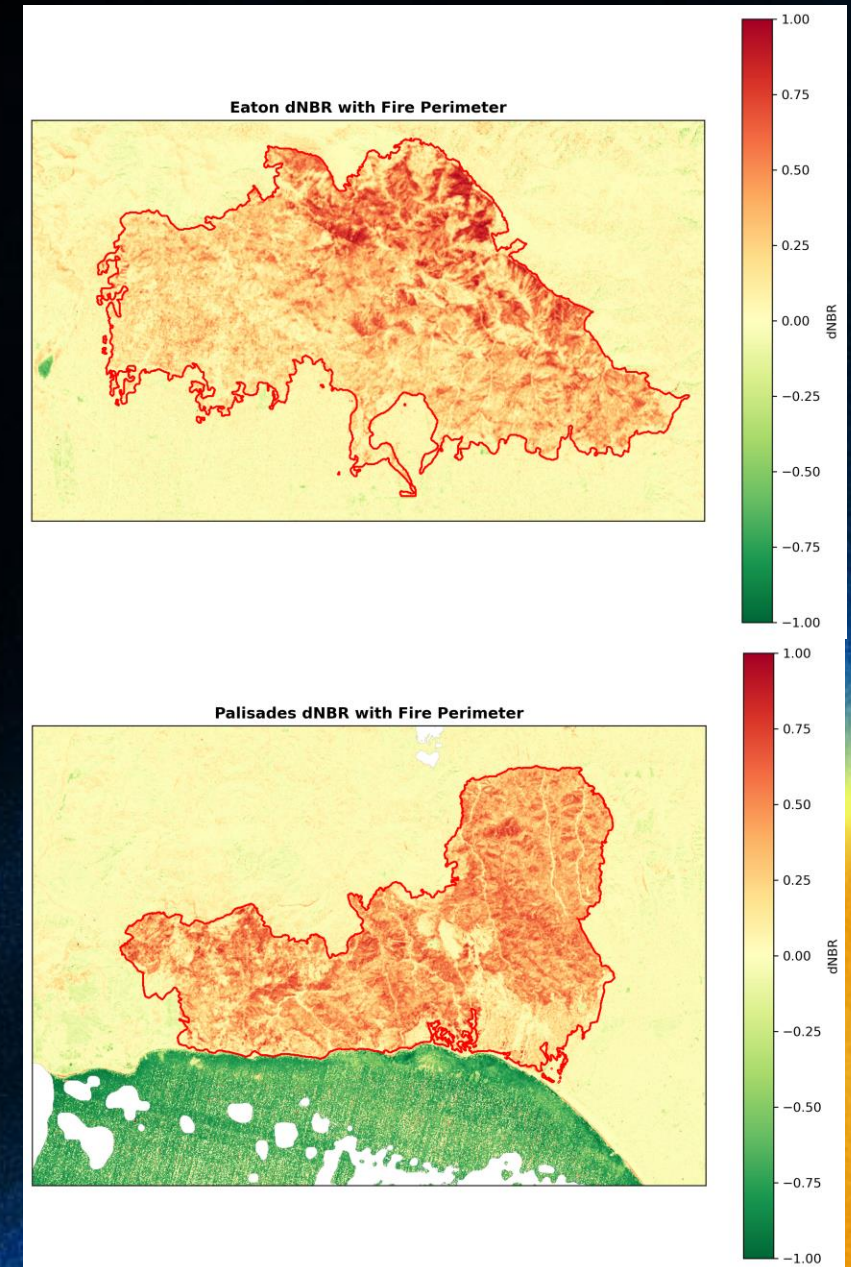
Event: 2025 Los Angeles Palisades and Eaton WUI Fires.

Problem: Wildland-Urban Interface (WUI) fires demand immediate, spatially explicit damage assessments to guide emergency response.

Current Issue: Traditional CAL FIRE Damage Inspection (DINS) takes weeks to complete and places inspectors in hazardous conditions.

Target Stakeholders: Los Angeles County Department of Regional Planning, local utility providers.

Objective: Generate a 20-meter resolution binary damage mask using Sentinel-2 imagery faster than field inspections allow.

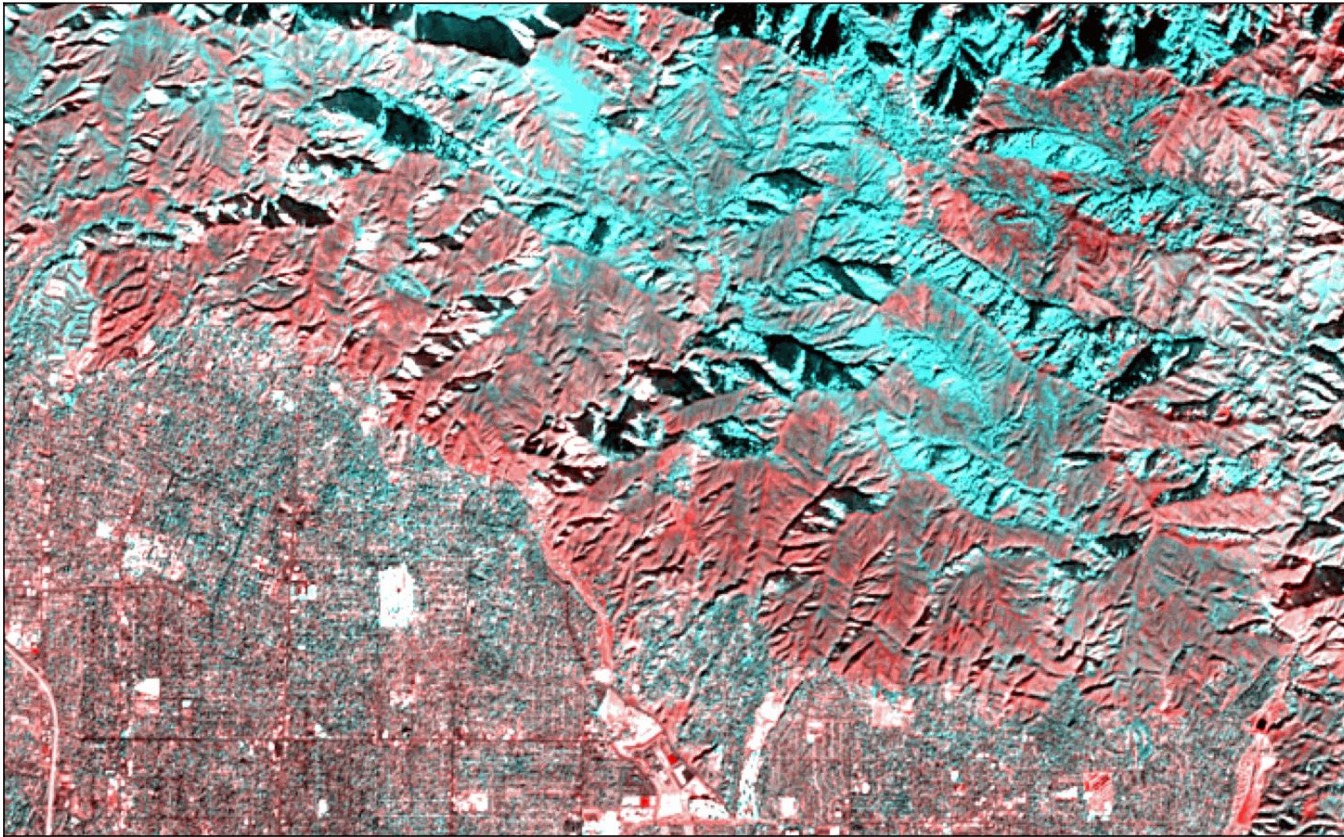


The background features a dark, almost black, central area. This is surrounded by a wide, glowing arc that transitions from a bright yellow on the left to a deep blue on the right. The glow has a soft, ethereal quality, with some internal texture and a slight gradient. The overall effect is reminiscent of a celestial body or a stylized horizon line.

DATA & METHODOLOGY

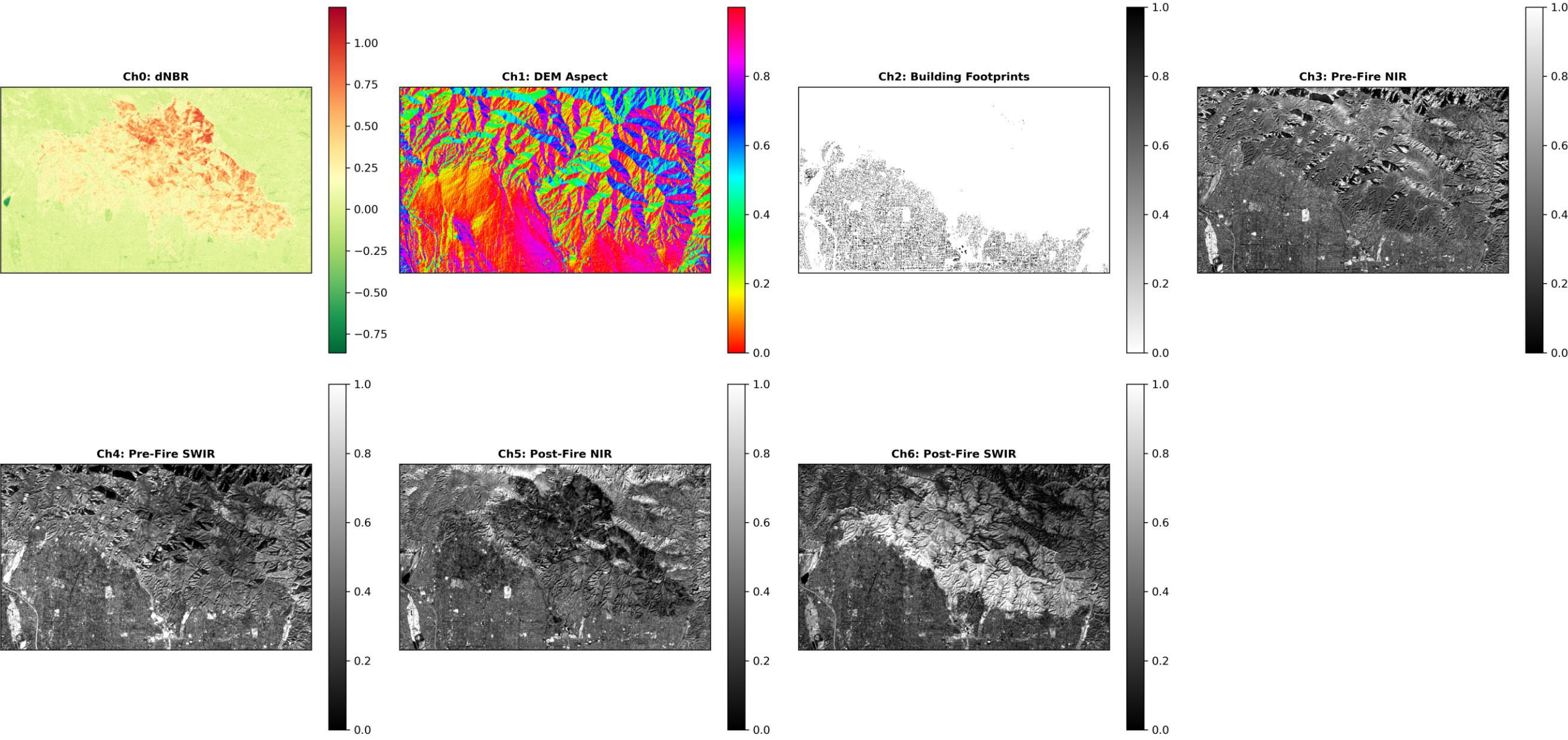
STACK

Stage 1: Pre-Fire SWIR False Color



- **Data:** Sentinel-2 L2A (NIR/SWIR for dNBR), USGS 3DEP DEM (aspect), and Microsoft Building Footprints.
- **Baseline Model:** Random Forest using pixel-wise classification.
- **Primary Model:** U-Net Deep Learning Architecture.
- **Rationale for Deep Learning:** U-Net skip connections preserve the fine-grained spatial boundaries necessary for the model to learn structural context rather than just spectral signatures.
- **Loss Function Optimization:** Binary Focal Cross-Entropy was implemented to aggressively penalize false negatives on the minority damaged class due to severe class imbalance.

Eaton Fire: 7-Channel Feature Tensor



The background features a dark, textured center surrounded by glowing, curved bands of light. A prominent yellow arc curves from the bottom left towards the right, while a blue arc is positioned just below it. The overall effect is reminiscent of a celestial body's horizon or a stylized representation of energy waves.

RESULTS & BENCHMARKS

QUANTITATIVE RESULTS VS. PROPOSAL BENCHMARKS

Pre-Defined Operational Benchmarks:

F1 > 0.50: Indicates a meaningful predictive signal.

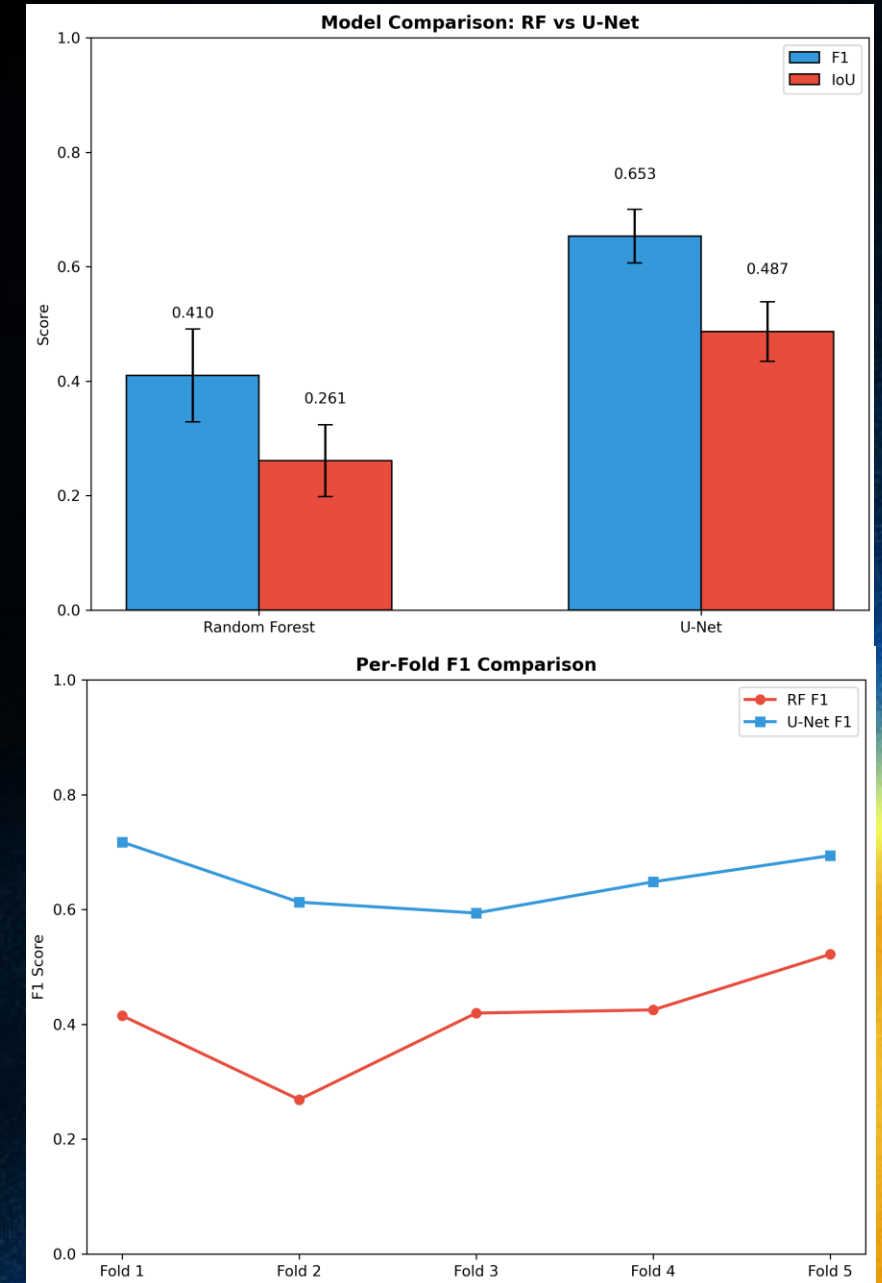
F1 > 0.70: General threshold for practical use.

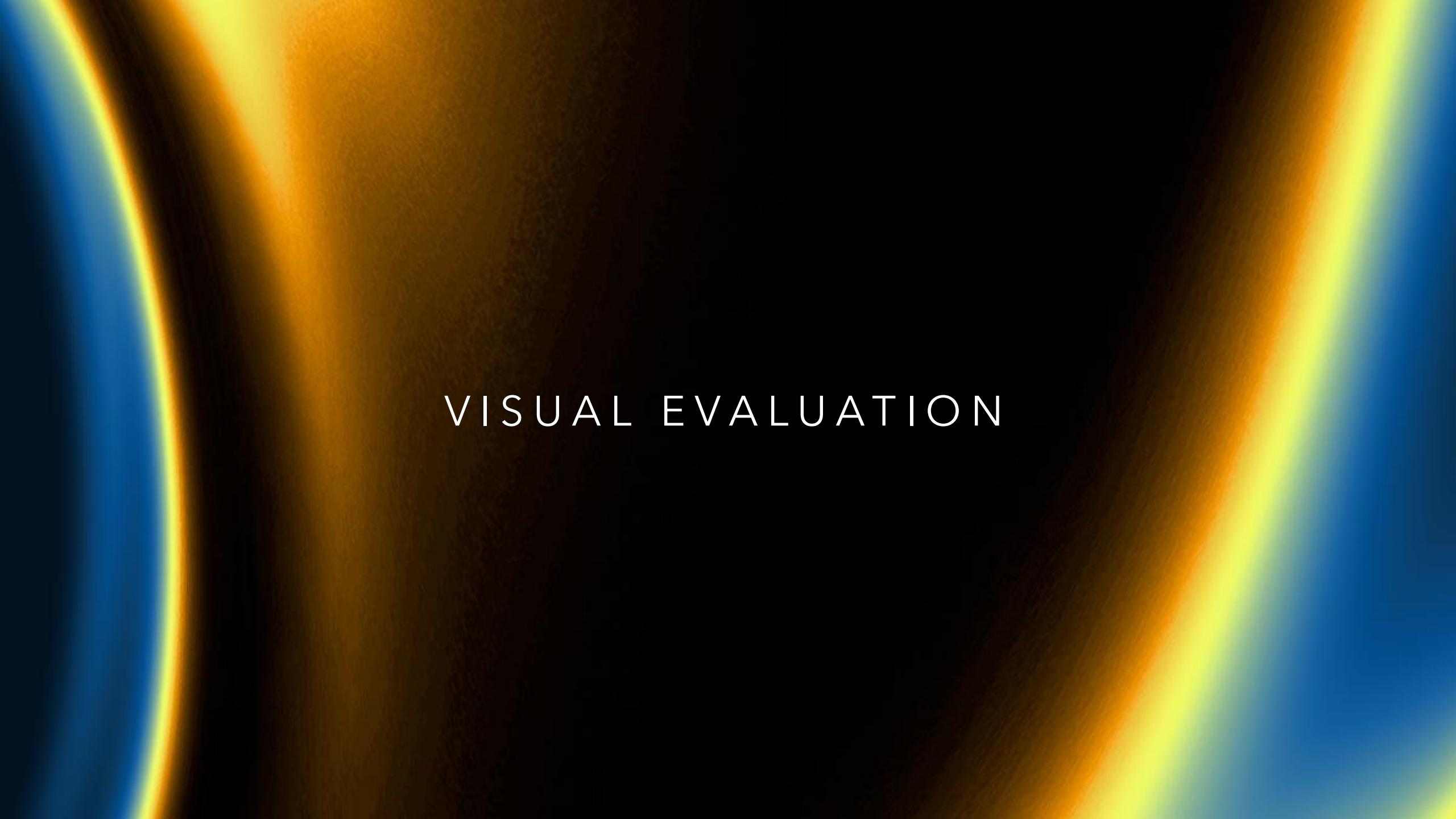
Evaluation Metrics (Damaged Class Only):

Random Forest Baseline: F1-Score of 0.41.

U-Net Primary Model: F1-Score of 0.65.

The deep learning architecture successfully extracted the spatial signal, drastically outperforming the baseline, but falls just short of the 0.70 fully autonomous deployment threshold.



The background features a black field with two prominent, curved, glowing bands. The band on the left is primarily blue with a bright yellow-orange inner edge. The band on the right is primarily orange with a bright yellow inner edge. Both bands curve from the top towards the bottom of the frame.

VISUAL EVALUATION

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Successes (True Positives):

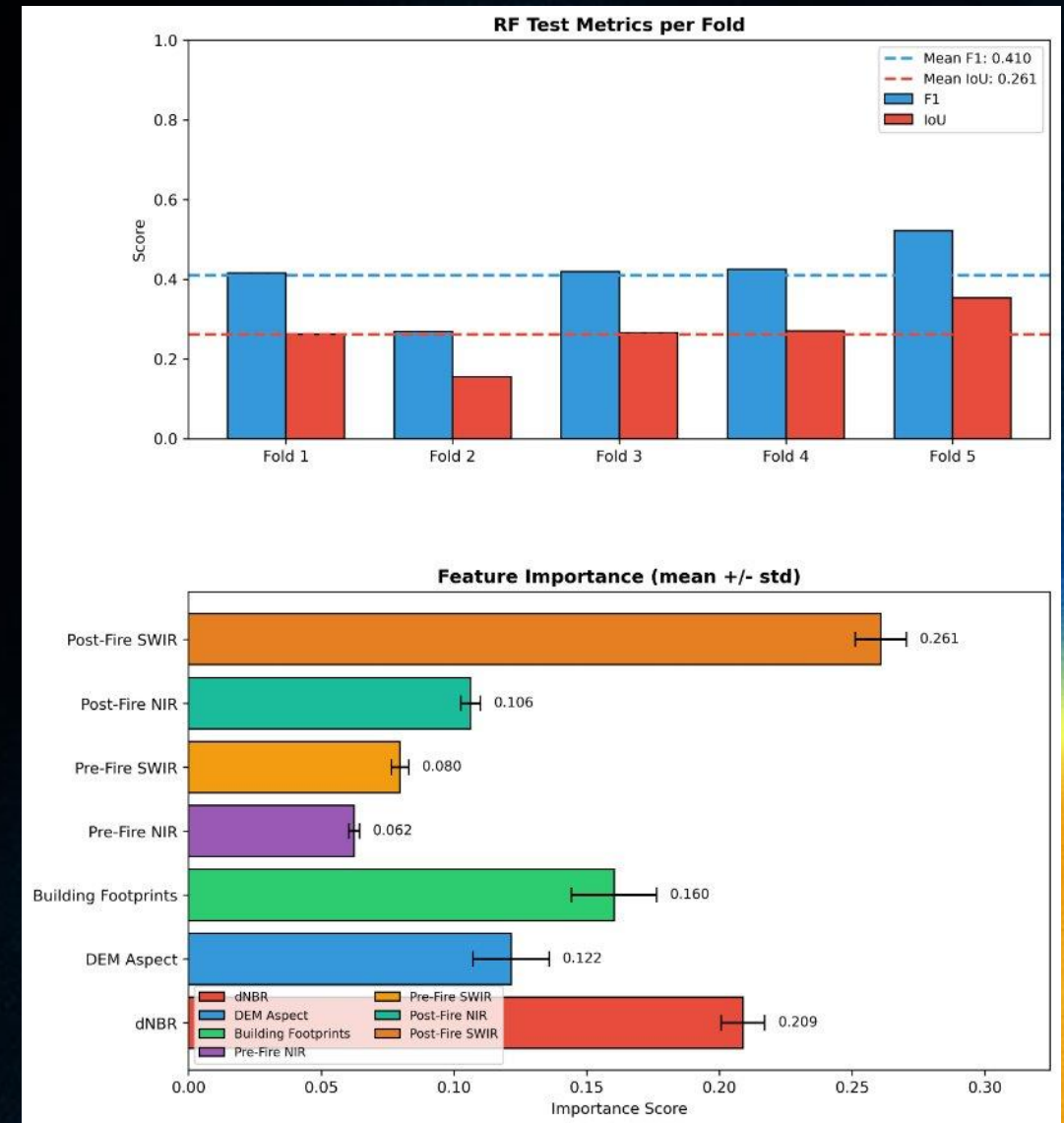
Highly effective at identifying dense, concentrated damage clusters.

Accurately maps zones where multiple adjacent structures were destroyed.

Failures (False Positives and Negatives):

Struggles to detect isolated, standalone damaged structures.

Conflates severe, high-intensity vegetation burns with actual structural destruction.

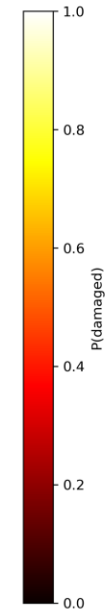
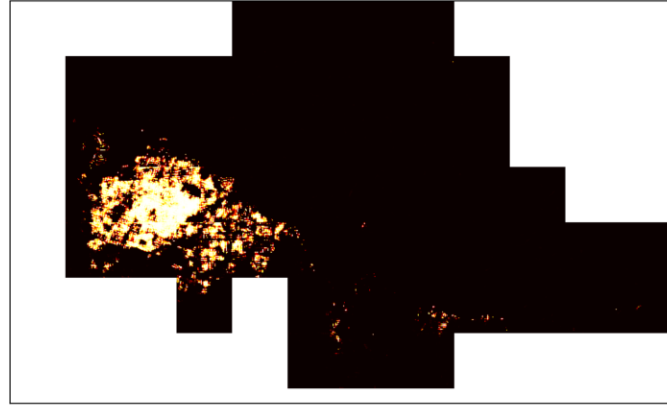


Eaton Fire: Spatial Prediction Map

Ground Truth (DINS Damage)



U-Net Prediction Probability

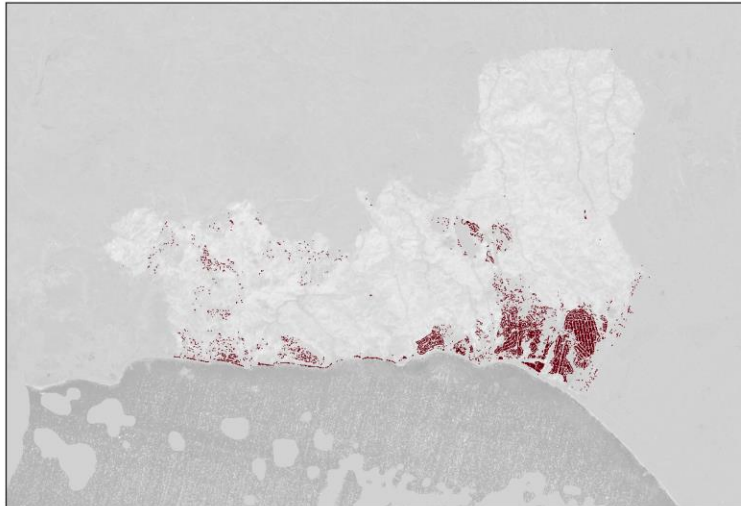


Spatial Error Map

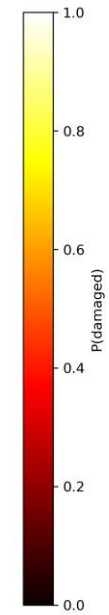
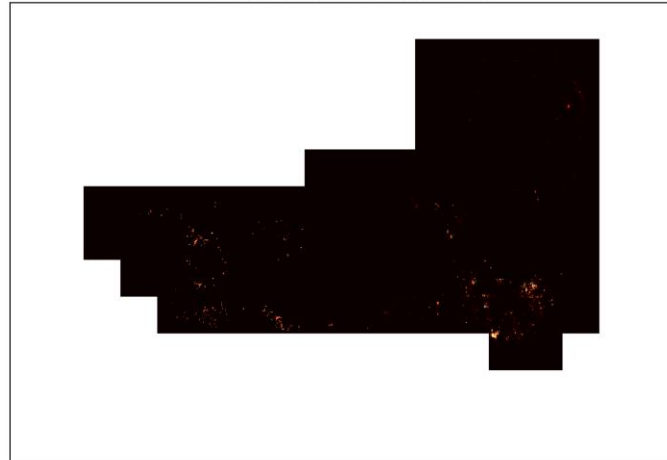


Palisades Fire: Spatial Prediction Map

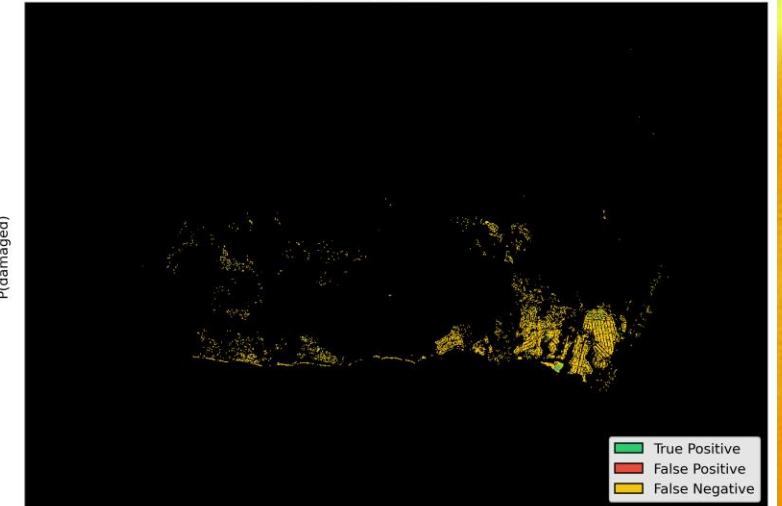
Ground Truth (DINS Damage)



U-Net Prediction Probability



Spatial Error Map





LIMITATIONS & IMPROVEMENTS

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Methodological Constraints:

Small sample size of 64x64 positive-class patches led to high fold-to-fold variance during cross-validation.

Sentinel-2's 20-meter resolution causes sub-pixel mixing, so individual structures cannot be cleanly differentiated.

Operational Improvements:

Current optical and SWIR indices fail if smoke or cloud cover persists immediately post-fire.

Solution: Integration of Sentinel-1 Synthetic Aperture Radar (SAR) backscatter and higher resolution satellite imagery.

Rationale: SAR penetrates smoke and is highly sensitive to the geometric structure of human-made buildings, which theoretically counters the false positives caused by vegetation burns. Higher resolution imagery will help with sub-pixel mixing and provide more granular detail for small structures.